DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACADEMIC ASSIGNMENT

CO4 - AT 2

INDUSTRY PROBLEM-SOLVING TASK

Network Topology Design for Real-World Scenarios

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Declaration

I hereby declare that this assignment is my own original work, prepared under the guidance of my course faculty, and has been submitted in accordance with the academic requirements of the course.

Question 1 - Star Topology for a 20-Person Startup

For a startup with around 20 employees, a star topology is the most suitable design because all computers, printers, VoIP phones, and other end devices are connected individually to one central switch. This structure is easy to install, manage, and troubleshoot, which is important for a growing office that may not have a large network support team.

Recommended devices and media: A managed Layer 2 or Layer 3 switch with at least 24 ports should be placed at the center of the network. The extra ports provide capacity for the existing users as well as future devices. Every workstation can be connected to the switch using Cat6 UTP cable. Cat6 supports Gigabit Ethernet and cable runs up to 100 metres, which is adequate for a normal office floor. The switch can uplink to a router/firewall that provides internet access, DHCP, and Network Address Translation. A wireless access point connected to the switch can extend Wi-Fi connectivity to laptops and mobile devices.

Reliability and management: The main advantage is fault isolation. If one workstation cable fails, only that particular node loses connectivity; the remaining employees continue to work. Since every connection terminates at the central switch, cable tracing and troubleshooting are also much simpler than in a shared-medium design. Moves, additions, and changes can be handled without disturbing the complete network.

Scalability and performance: A 24-port managed switch gives the startup room to add more systems and also allows centralized monitoring and control. Dedicated switch ports reduce unnecessary interference between users and provide a clean structure for future upgrades. Wireless users can also be supported without changing the wired topology.

Conclusion: The proposed star topology combines low complexity, reliable fault isolation, Gigabit-capable Cat6 cabling, centralized administration, and easy expansion. Hence, it is a practical and industry-relevant network design for a 20-person startup.

Question 2 - Mesh Topology for a Bank Requiring Near-Zero Downtime

A bank requires continuous availability because a network outage can interrupt online transactions, ATM services, internal applications, and communication with customers. For this reason, a full or partial mesh topology is appropriate at the critical network layers. Mesh design provides multiple communication paths so that the failure of one link or device does not immediately stop the entire network.

Topology design: At the core, important devices such as routers, core switches, servers, and firewalls should have redundant connections. A full mesh can be used where the number of critical devices is small. In a larger deployment, a partial mesh is more practical: core switches and firewalls are interconnected, while distribution and access devices use redundant uplinks to at least two core switches.

Devices and transmission media: Dual enterprise-grade routers, such as Cisco ASR or Juniper MX class devices, can provide redundant internet connectivity using BGP. Links between core devices should use fiber optic cabling. Single-mode fiber is suitable for long distances between buildings, while multimode OM4 fiber can be used within a data center. Fiber supports very high bandwidth, low-latency communication, and is immune to electromagnetic interference.

Fault tolerance: If one path fails, traffic can be redirected through another available path using routing protocols such as OSPF or BGP. This ability to reroute traffic in a very short time is the key reason mesh topology is preferred for critical financial environments. Redundant power supplies, UPS systems, and RAID storage further strengthen service continuity.

Conclusion: Although mesh networking requires more links, ports, and careful administration, the higher cost is justified by the bank's need for availability, integrity, and reliable transaction processing. A partial mesh at the core gives a strong balance between redundancy and practical implementation.

Question 3 - Low-Cost Network for a Small Classroom Lab

For a small classroom laboratory where the main constraint is budget, a bus topology offers a simple and low-cost design. In a traditional bus network, all computers share a single communication backbone, so the amount of cabling and networking hardware required is less than in more complex topologies. This makes the approach suitable for a small supervised environment with light network usage.

Traditional implementation: A classic bus topology can use RG-58, 50-ohm coaxial cable as the common backbone. Each computer connects to the cable using a T-connector and BNC connector. A 50-ohm terminator must be placed at both ends of the backbone to absorb signals and prevent reflections that could interfere with communication.

Modern low-cost alternative: For current equipment, an inexpensive 8-port or 16-port unmanaged switch and Cat5e or Cat6 UTP patch cables can be used to achieve a similarly simple low-cost classroom arrangement. The teacher's PC can provide shared files or resources through built-in Windows or Linux file-sharing features, so a dedicated server is not necessary.

Limitations and suitability: The major weakness of a true bus network is that every device depends on the same backbone. A break in the main cable can disrupt all connected systems. Because the medium is shared, collisions and reduced performance can also occur as more devices communicate at the same time. Troubleshooting a backbone fault may be more difficult than locating a failed link in a star network.

Conclusion: For a small lab used mainly for internet browsing, demonstrations, and basic file sharing, the reduced cost and simple structure can outweigh these limitations. The design is therefore acceptable when the number of computers is limited and the network is monitored by the instructor.

Question 4 - Hybrid Topology for a Large University Campus

A large university contains lecture halls, laboratories, libraries, hostels, administrative blocks, and thousands of users. A single topology cannot efficiently satisfy all these requirements. Therefore, a hybrid topology is appropriate because it combines different designs at different network layers to achieve redundancy, manageability, and campus-wide scalability.

Campus backbone: At the core level, high-speed fiber optic links should interconnect core routers and multilayer switches in a ring or partial mesh arrangement. Single-mode fiber operating at 10 Gbps or 40 Gbps can be used between major buildings. The redundant backbone ensures that the failure of one link does not isolate a complete building from the rest of the campus.

Building-level design: Each building should have a distribution switch connected to the campus core through dual fiber uplinks. Inside the building, a star topology can be used from the distribution switch to floor-level access switches. End devices such as desktop computers, printers, and laboratory systems can then connect through Cat6 cabling.

Hierarchical architecture: This creates the common three-layer model of core, distribution, and access. The structure improves organization and makes expansion easier. If a new building is added, a new distribution switch can be installed and connected to the core without redesigning the existing campus network. Wireless access points can connect to access switches using Power over Ethernet, reducing the need for separate electrical cabling at each access-point location.

Conclusion: The hybrid design provides redundancy at the backbone, simple star-based management within buildings, and flexible growth across the campus. It therefore supports large numbers of users while remaining easier to expand and maintain than a single uniform topology.

Question 5 - Structured Network for Multiple Company Departments

A company with departments such as HR, Finance, Sales, and IT requires a structured network that allows internal communication while keeping departmental traffic organized and secure. A hierarchical design using Layer 2 access switches, a central Layer 3 switch or router, VLANs, and an internet router/firewall provides a scalable solution.

Access layer: Each department should be connected to its own Layer 2 switch. Computers, printers, and other devices within the same department can communicate through this switch. The physical access switches provide a clear point for connecting and managing end-user devices.

VLAN segmentation and routing: A separate Virtual LAN should be configured for each department. VLANs logically isolate departmental traffic, reduce the size of broadcast domains, and improve security. The access switches can be connected to a central Layer 3 switch or router, which performs inter-VLAN routing whenever authorized communication is required between departments.

Internet access and reliability: A router connects the internal company network to external networks such as the internet, and firewall functionality can be used to control traffic entering or leaving the organization. Proper IP addressing and subnetting should be applied to each VLAN so that the network remains easy to identify, troubleshoot, and manage. Redundant links or backup paths may be added for important systems to increase reliability.

Scalability and security: The hierarchical arrangement allows new users or departments to be added by introducing additional access switches and VLANs without redesigning the full network. Logical separation also prevents unnecessary traffic from one department from reaching every other department and makes access policies easier to enforce.

Conclusion: This design combines switching, VLAN-based separation, inter-VLAN routing, firewall protection, subnetting, and optional redundancy. It improves performance and security while giving the organization a manageable network that can expand as the company grows.